Meghal Dani

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Research Interest

Multi-modal learning (text, biosignals, medical images, videos), Safe AI in healthcare, Large Language Models (LLMs)

Education

Ph.D. in Computer Science — Aug. 2022 – Present
International Max Planck Research School for Intelligent Systems, University of Tuebingen

Masters of Technology in Computational Biology — Jul. 2017 – Jul. 2019
Indraprastha Institute of Information Technology Delhi (IIIT-D) — *GPA: 9.55/10 (Distinction)*

Bachelor of Engineering in Computer Science — Jul. 2012 – May. 2016
Birla Institute of Technology (BIT, Mesra) — *GPA: 8.47/10 (Distinction)*

Research and Industry Experience

Visiting Researcher, Neuroscience Institute, Groote Schuur Hospital — Nov. 2025 – Dec. 2025
Cape Town, South Africa

- Advisors: Dr. MV Gule and Dr. rer. nat. Stefanie Liebe
- Developing AI pipelines using LLMs and VLMs to extract seizure types and onset zones from handwritten epilepsy records, uncovering cross-cultural variations in symptoms and clinical workflows across South African cohorts.

Researcher, TCS Research and Innovation Labs — Aug. 2019 – Aug. 2021
New Delhi, India

- Advisors: Dr. Lovekesh Vig, Ramya Hebbalaguppe
- Developed anchor-free universal lesion detection network for CT scans achieving 86.05% sensitivity, published at BMVC'22 and ISBI'22 with a patent
- Created lightweight 3D pose estimation using skeletal graphs with 11x space and 3x time reduction, published at SIGGRAPH'20 and WACV'21 with a patent

Graduate Student Researcher, Image Analysis and Biometrics Lab (IAB) — Aug. 2018 – Jul. 2019
New Delhi, India

- Advisors: Prof. Dr. Richa Singh and Prof. Dr. Mayank Vatsa
- Developed ML framework for autism detection from rs-fMRI (72.5% accuracy; +6% on SOTA) and identified camouflaging patterns in female ASD cohort via self-awareness network connectivity analysis
- Nominated for best thesis award

Research Intern – TCS Research and Innovation Labs — May 2018 – Jul. 2018
New Delhi, India

- Advisors: Ramya Hebbalaguppe
- Developed markerless fingertip tracking system for 3D gesture interaction in mixed reality; published at ISMAR'18.

Publications and Patents

- "Parameter-Efficient Self-Supervised Adaptation for EEG-FM under Fixed Computational Budgets."
  M. Dani, S. Liebe.
  under review at MICCAI 2026
- "SemioLLM: Evaluating Large Language Models for Diagnostic Reasoning from Unstructured Clinical Narratives in Epilepsy."
  M. Dani, M. J. Prakash, Z. Akata, and S. Liebe.
  Nature Communications Medicine 2026 (Also presented at ICML AI for Science workshop 2024)
- "Towards Automatized Analysis of Epileptic Seizure Behavior in VEEG Recordings."
  M. Dani, M. J. Prakash, Z. Akata, and S. Liebe.
  Bernstein Conference 2023

- "Devil: Decoding vision features into language."
  M. Dani*, I. Rio-Torto*, S. Alaniz, and Z. Akata.
  *DAGM GCPR 2023 (**Oral**), (Also presented at ICCV-CLVL workshop 2023*
- "An efficient anchor-free universal lesion detection in CT-scans."
  M. Sheoran*, M. Dani*, M. Sharma, and L. Vig.
  International Symposium on Biomedical Imaging (ISBI), 2022
- "DKMA-ULD: Domain knowledge augmented multi-head attention based robust universal lesion detection."
  M. Dani*, M. Sheoran*, M. Sharma, and L. Vig.
  The British Machine Vision Conference (BMVC) 2022
- "3DPoseLite: A compact 3d pose estimation using node embeddings."
  M. Dani, K. Narain, and R. Hebbalaguppe.
  The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2021
- "PoseFromGraph: Compact 3-d pose estimation using graphs."
  M. Dani, A. Popli, and R. Hebbalaguppe.
  SIGGRAPH, 2020

Patents

- "Method and system for domain knowledge augmented multi-head attention based robust universal lesion detection."
  M. Sheoran*, M. Dani*, M. Sharma, and L. Vig.
  U.S. Patent 12131466, Oct. 2024
- "Methods and systems for generating end-to-end model to estimate 3-dimensional (3-D) pose of object."
  R. Hebbalaguppe, M. Dani, and A. Popli.
  U.S. Patent 12033352, Jul. 2024

Relevant Projects

Brain Research International Data Governance & Exchange (BRIDGE) 2025
Technologies Used: MRI data preprocessing, Neuroimaging QC, Statistical analysis

- Supported development of international data governance frameworks by systematically accessing, preprocessing, and analyzing multi-country brain MRI datasets (SUDMEX_CONN and Nigerian Brains) while documenting legal, technical, and training barriers to cross-border data sharing.

Seizure Detection Challenge (SzCORE, EPFL) 2025
Technologies Used: PyTorch, State-Space Models (S4, Hyena), EEG Segmentation

- Developed S4 and Hyena-based state-space model architectures for EEG-based seizure detection and segmentation, achieving 3rd position out of 30 submissions in the SzCORE epilepsy benchmark challenge.

Towards Automatized Analysis of Epileptic Seizure Behavior in VEEG Recordings 2024
Technologies Used: PyTorch, OpenCV, FFmpeg, MediaPy, Computer Vision

- Developed a pose-estimation framework using TAPIR that tracks 21 body joints from a single camera view in low-light, low-resolution hospital videos with heavy occlusions (blankets, staff, equipment), generating joint pose trajectories to detect seizure vs non-seizure events and classify seizure type.

Past Achievements, Awards and Community Service

- **Speaker**: at Loops and SFB "Robust Vision" women's retreat 2025
- Secured third position on **epilepsy benchmark challenge** to detect seizures from EEG data held by EPFL among 30 participating teams (2025)
- Soft Skills Seminar Series (S4) Workshop Organizer at IMPRS-IS (2024 - Present)
- **Ph.D. Representative** in Search Committee for Tenure-Track Professor of Machine Learning and Intelligent Systems, University of Tuebingen (2024)
- Selected to attend **OxML MLxHealth** 2024, organized by Oxford University's Deep Medicine Program, CIFAR, and AI for Global Goals
- **Awarded ClinBrAIn PhD Fellowship**: by EKFS, Tuebingen, Germany (2022)
- **Best Thesis Award Nominee**: for masters thesis at IIIT-Delhi (2019)
- **Travel Award** from IIIT-D to present at International Symposium on Mixed and Augmented Reality (ISMAR) (2018)
- **Postgraduate Fellowship**: Received funding from the Government of India (2017)
- **Academic Reviewer**: ICCV 2023 - 2025, CVPR 2025, ICML 2024, MICCAI 2024 - 2026, ACCV 2024